

NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Fifth Semester B.E. (Civil) (Credit System) Degree Examinations
January 2023

20CV501 – STRUCTURAL ANALYSIS - II

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **each Unit**.*

Unit – I

Marks BT* CO* PO*

1. Determine the forces in all the members of the truss shown in Fig.Q1 using Minimum Strain Energy method. All members have same material and cross-sectional area.

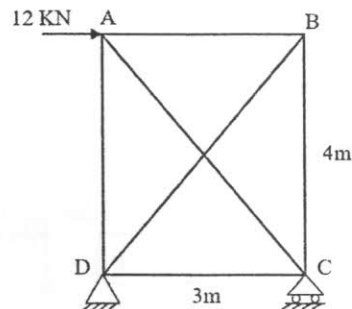


Fig.Q1

20 L*3 1 1, 2

2. In the equilateral triangular truss shown in Fig.Q2, the cross-sectional area of all members is 3000mm² each. Find the amount of error in length of member AB, if the member force in AB is 30kN. Also find forces in all other members. Take $E = 218 \text{ kN/mm}^2$

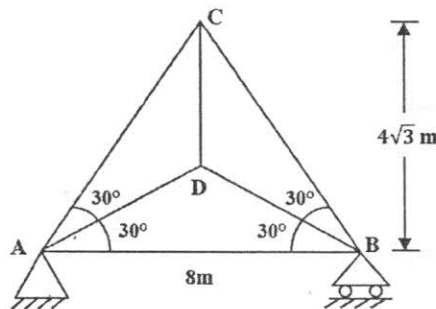


Fig.Q2

20 L2 1 1, 2

Unit – II

3. Analyze the continuous beam ABCD shown in Fig. Q3 by slope deflection method. Draw BMD, SFD and elastic curve.

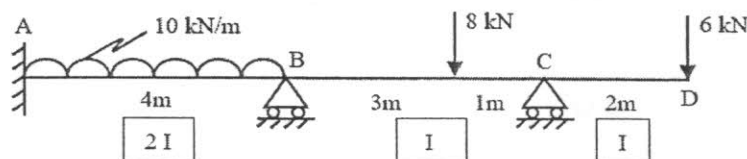


Fig.Q3

20 L2 2 1, 2

4. Analyze the frame shown in Fig.Q4 by slope deflection method. Draw BMD and elastic curve.

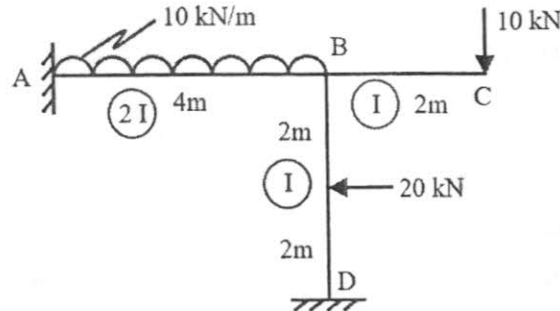


Fig.Q4

20 L2 2 1, 2

Unit – III

5. Analyze the continuous beam ABCD shown in Fig.Q5 by moment distribution method. Draw BMD, SFD and elastic curve.

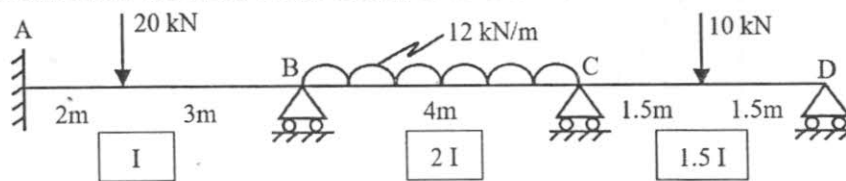


Fig.Q5

20 L3 3 1, 2

6. Analyze the frame shown in Fig.Q6 by moment distribution method. Draw BMD and elastic curve.

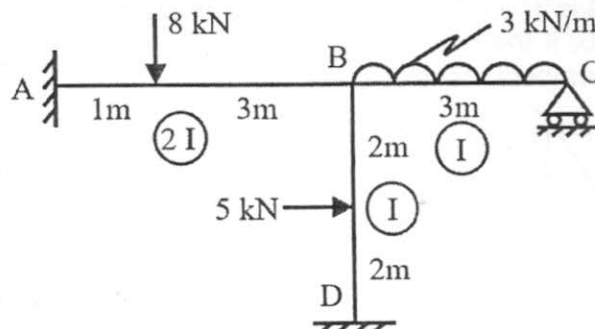


Fig.Q6

20 L3 3 1, 2

Unit – IV

7. Analyze the continuous beam ABCD shown in Fig. Q7 by Kani's method. Draw BMD and elastic curve.

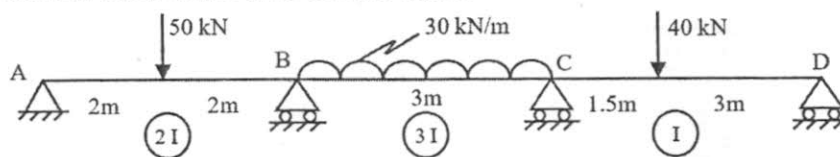


Fig.Q7

20 L2 4 1, 2

8. Analyze the frame shown in Fig.Q8 by Kani's method using symmetry of the frame and draw BMD. Assume I as constant.

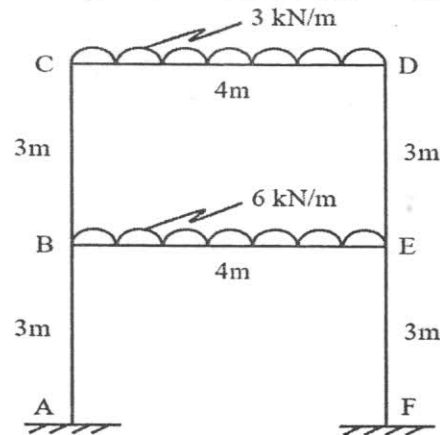


Fig.Q8

20 L2 4 1, 2

Unit – V

9. Analyze the continuous beam shown in the Fig.Q9 by flexibility matrix method and draw BMD.

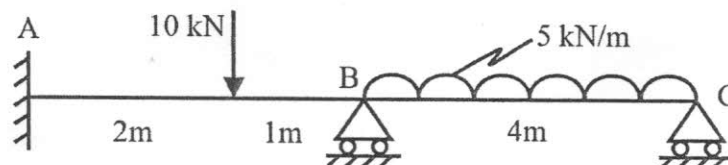


Fig.Q9

20 L3 5 1, 2

10. Analyze the continuous beam shown in the Fig.Q10 by stiffness matrix method and draw BMD.

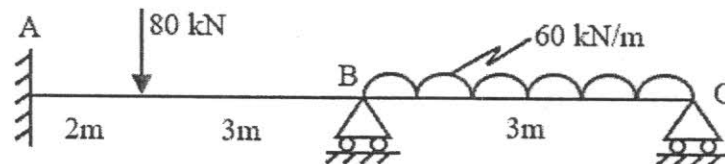


Fig.Q10

20 L3 5 1, 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
